

Case Study 14 — Calibration Block

Linear energy map, oscillator scaling, and per-band width overrides

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1 Case Study 14 — Calibration Block

Apply a TOML calibration before broadening. The calibration block can shift TD-DFT excitation energies with a linear map $E_{\text{cal}} = aE + b$, scale oscillator strengths $f_{\text{cal}} = \alpha f$, and override the broadening width or reorganization energy per excited state.

The calibration is applied **before** broadening. Wavelength is recomputed from the calibrated energy via $\lambda = 1239.841984 \text{ eV nm} / E_{\text{cal}}$ only when the linear energy map is non-identity. The identity calibration ($a = 1$, $b = 0$, $\alpha = 1$, no band overrides) produces bit-identical output to running without `--calibration`.

The user-selectable knob is:

- CLI: `--calibration PATH` (path to a TOML file)
- Library: `calibration=Calibration(...)` parameter of `analyze()`

Note. The HTTP API does not accept calibration files. Use the CLI or the Python library for this feature.

1.1 TOML schema

```
[energy]
a = 1.0    # multiplicative factor (default 1.0 = identity)
b = 0.0    # additive shift in eV (default 0.0)

[oscillator]
alpha = 1.0 # scaling factor for all oscillator strengths

[[band]]
index = 1      # 1-based excited-state index
sigma_ev = 0.25 # width override in eV (optional)
```

```
reorganization_ev = 0.40 # Marcus-Hush reorganization energy (optional)
```

1.2 What you will produce

Folder	By	Contents
output/cli/tables/	CLI	Result tables (CSV / JSON / XLSX)
output/cli/plots/	CLI	TIFF plots per sample, light source, and (sample, light) pair

Light sources used: **AM1.5G + LED-B4**.

1.3 1. Prerequisites

If you have never run any case study before, follow [Case Study 01 §1 Prerequisites](#) once to install the package and either the Python or Docker option for the HTTP API.

1.4 2. Inputs

The case-study `input.json` lists 2 Gaussian TD-DFT outputs. The case study also includes a `calib.toml` file that leaves energy and oscillator scaling at their identity values ($a = 1$, $b = 0$, $\alpha = 1$) and overrides the width of the first excited state to $\sigma = 0.25$ eV.

1.5 3. Run with the CLI

```
overlap-calculator analyze --input case_studies/14_calibration_block/input.json --out
↪ case_studies/14_calibration_block/output/cli --default-light-sources AM15G,LEDB4 --calibration
↪ case_studies/14_calibration_block/calib.toml --plot-dpi 400
```

The `run_manifest.json` at the output root records the full calibration block under `parameters.calibration`.

1.6 4. HTTP API

The HTTP API does not accept calibration files. Use the CLI or the Python library for this feature.

1.7 5. Drive the CLI from one Python script

```
python case_studies/14_calibration_block/submit.py
```

1.8 6. Expected output layout

```
case_studies/14_calibration_block/output/
+-- cli/
+-- run_manifest.json
+-- tables/results.{csv,json,xlsx}          (2 samples x 2 light sources = 4 rows)
+-- tables/results_timings.{csv,json,xlsx}
+-- tables/descriptor_summary.{csv,xlsx}
+-- tables/ranking_by_light_source__<metric>.{csv,json,xlsx}
+-- tables/ranking_by_sample__<metric>.{csv,json,xlsx}
+-- plots/
```

Compare `gaussian_lambda_max_nm` between the calibrated and the default run: the -0.05 eV shift should produce a red-shift in the peak wavelength.

1.9 7. Related case studies

- [Case Study 08 — Broadening Sigma](#)
- [Case Study 13 — Marcus-Hush Width](#)
- [Case Study 15 — Vibronic Tabular Branch](#)